## Statistical Comparison of Algorithms

To compare the algorithms, the **Friedman Test** was applied, followed by the **Nemenyi Test** at a significance level of **α = 0.05**.

Experiment parameters:

- **Number of algorithms (k):** 3
- **Number of datasets (N):** 4
- **Critical value (qα):** 3.3145
- **Critical Difference (CD):** 2.3437

### Comparison Matrix

| | Dataset | svm_lin | tree | knn |
|---|---|---|---|---|
| **0** | D1 | 1.000000 | 0.990667 | 1.000000 |
| **1** | D2 | 0.881333 | 0.828667 | 0.860667 |
| **2** | D3 | 0.666667 | 0.900333 | 0.920333 |
| **3** | D4 | 0.521333 | 0.958333 | 0.958667 |

### Accuracy Matrix

| | svm_lin | tree | knn |
|---|---|---|---|
| **0** | 1.000000 | 0.990667 | 1.000000 |
| **1** | 0.881333 | 0.828667 | 0.860667 |
| **2** | 0.666667 | 0.900333 | 0.920333 |
| **3** | 0.521333 | 0.958333 | 0.958667 |

### Rank Matrix

Algorithms were ranked for each dataset based on accuracy (**rank 1 = best performance**). In cases where two algorithms achieved the same accuracy, **average ranks were assigned** (1.5).

| | svm_lin | tree | knn |
|---|---|---|---|
| **0** | 1.5 | 3.0 | 1.5 |
| **1** | 1.0 | 3.0 | 2.0 |
| **2** | 3.0 | 2.0 | 1.0 |
| **3** | 3.0 | 2.0 | 1.0 |

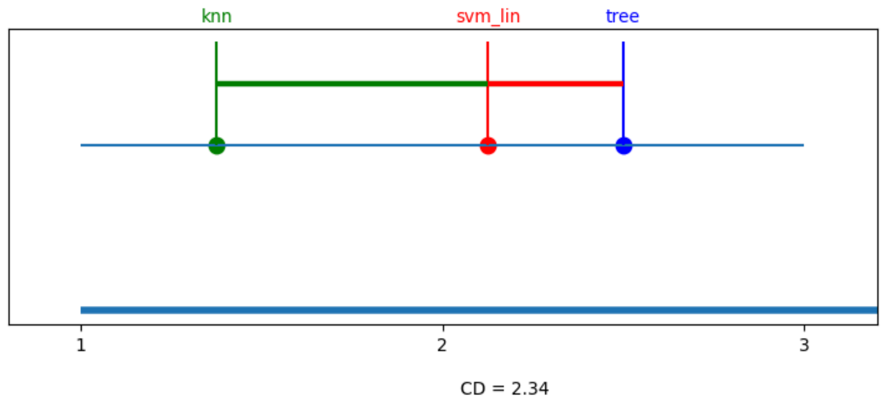
### Average Ranks

Average ranks were computed across the datasets. Lower rank means better performance.

| | |
|---|---|
| **svm_lin** | 2.125 |
| **tree** | 2.500 |
| **knn** | 1.375 |

### Critical Difference Diagram

In the diagram, algorithms on the **left have better average rank**




### Conclusion

From the CD diagram:

- **KNN has the best average rank**
- **Decision Tree has the worst average rank**
- **SVM lies between them**

However, since the differences between the ranks are **smaller than the Critical Difference (CD = 2.3437)**, **no statistically significant difference is observed between the algorithms at α = 0.05**.